

How Should a Home Robot Update Its Memories When Human Routines Change?

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Abstract: Robots deployed in homes should keep learning about the specific home they work in. A robot that assists with chores should keep track of where objects are, but people move objects according to their own routines and habits. One approach is to have a language model maintain a written memory of the home from the robot’s own observations and read that memory when deciding where to search. Prior evaluations of such memories use static worlds where the correct answer never changes. Whether a written memory can handle a household whose routine changes and then changes back is not well tested. We study this in simulated households where a resident falls ill, recovers, and later falls ill again. The distribution of object locations during illness differs from the distribution during normal routine. The robot sees only the rooms it opens and is never told that anyone is ill. We compare a simple rule that searches where an object was last seen, several language-model memories that store only written summaries, and the same summaries paired with the raw record of sightings. The simple rule outperforms every summary-only memory, and adding the raw record closes that gap. The return of the illness reveals the larger difference. All methods drop in accuracy when the second illness begins, but methods with the raw record are less wrong than they were at the first illness, while summary-only methods are about as wrong as the first time. Their summaries record where objects moved but not why. When we tell the model the cause directly, it predicts the new locations correctly for a full day, then discards the explanation in its next update after one contrary observation. Self-summarizing text memory did not reliably turn the robot’s experience into reusable knowledge. Memories for changing homes should keep raw observations alongside summaries and represent the state of the household separately from facts about objects, and how a memory applies past experience to a situation that returns should be an evaluation criterion for robot memory.

1 Introduction

Robots deployed in homes must keep learning about the specific home they work in. A robot that helps with chores needs to know where objects are, and where objects are depends on the people who live there and on what those people are doing that week. A laptop that lives on the office desk may spend a sick week on the bedroom nightstand and return to the desk after.

One way to keep up is to give the robot a language model that maintains a written memory of the home. Each night it reads what the robot saw and revises the memory, and each day the robot reads the memory when deciding where to search [1, 2, 3]. Robot foundation models are starting to include a layer like this: MEM pairs raw recent observations with a long-term memory written in natural language [4]. The appeal is that the memory could notice that several objects moved for one shared reason, or recognize a situation it has seen before and act on it, without anyone labeling what changed.

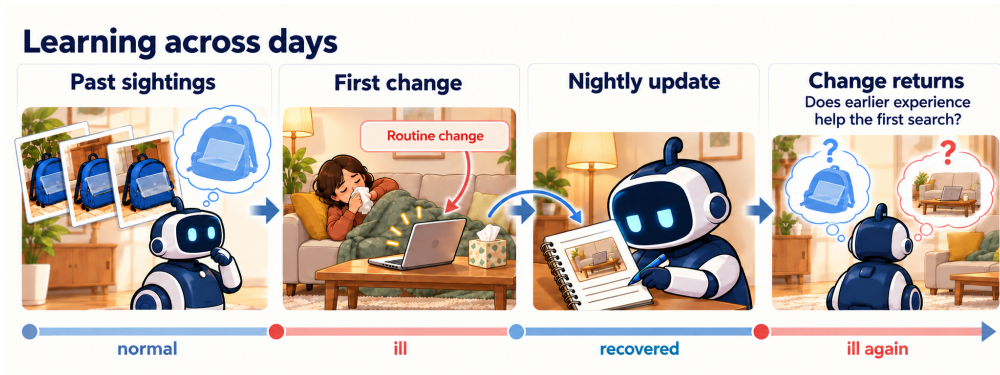


Figure 1: **Learning from searches across a recurring change.** The robot gathers sightings only from rooms it opens, updates its memory each night, and reads that memory to choose rooms the next day. The resident’s illness is hidden from the robot. When it returns, the question is whether earlier experience improves the first search.

Recent work gives reason for caution. Zhang et al. [5] show that when a language model repeatedly rewrites its own memory, the memory degrades even when the world is static and every episode it learns from is complete and correct. Simply keeping the raw episodes is competitive with every consolidation method they test. Robot memory benchmarks point at a different limit. RoboMME, RoboMemArena and RoboMME-Interference evaluate recall within a single manipulation episode lasting minutes [6, 7, 8], and spatio-temporal memories such as ReMEMbR and Khronos answer where an object was seen or map objects that moved while the robot was away [9, 10]. None of these tests a memory that a robot writes for itself over weeks, from observations it chose to collect, in a home whose routine changes and then changes back.

That last case is where written memory should matter most and where it is hardest. In a static world, a memory only has to avoid corrupting what it knows. In a changing home, it must revise what has gone stale and keep what will be needed again, and it must do both from a handful of room openings a day, with the evidence for a change spread across more days than one nightly update covers. When the change returns, a memory that learned should do better than it did the first time. A memory that only tracked recent observations will not.

We study this in simulated households where one resident falls ill, recovers, and later falls ill again. Object locations during illness follow a different distribution from the normal routine. The robot searches for objects, sees only the rooms it opens, and is never told that anyone is ill. We compare a simple rule that searches where an object was last seen, several language-model memories that store only written summaries, and the same summaries paired with the raw record of sightings. We find:

- **The last-seen rule outperforms every summary-only memory**, before and after the illness, across ten households.
- **Giving the same summaries the raw sighting record closes the gap.** The gain is recovery from a wrong first guess, not a better first guess.
- **When the illness returns, memories with the raw record are less wrong than at the first illness.** Summary-only memories are as wrong as the first time.
- **The summaries record where objects moved but never why.** Given the cause in one sentence, the model predicts the moved locations correctly for a day and discards the explanation in its next update.

2 Setting and Methods

Households and the change in routine. Each simulated household has two or three residents, eight or nine rooms with about forty places between them, and roughly 80 to 100 objects that res-

idents move as they go about their day. Days 1–13 follow the normal routine. On days 14–23 one resident is unwell and stays home, which changes where their objects tend to be: a glass becomes more likely to be on the bedroom nightstand than in the kitchen. Days 24–31 return to normal. The 50-day run repeats the illness on the same resident on days 32–41 and returns to normal on days 42–49. The second illness is a fresh draw from the same distribution, not a replay. The robot is never told about the illness, except in the diagnostic arm of Section 3.3. Full details and statistics are in Appendix A.

Questions and searches. Each day the robot receives object-location questions spread across the day, on the same schedule for every method. For each question it opens a room, and up to two more if the object is absent. Every opening records every object in the room as a sighting with its spot and time, so the robot also learns about objects it was not asked about. There is no separate patrol. We score *first room right* and *found within three rooms*. *Moved objects* are those whose most common daytime spot differs between normal and illness days, read from the simulator’s ground truth. We use three runs and never pool them: ten households at 8 questions a day for 31 days; three households at 24 a day for 50 days, with the second illness; and five other households at 4, 8 and 24 a day, to vary how much the robot observes.

Methods. All language-model methods use Qwen3.8–27B [11] at temperature zero with thinking disabled and a JSON schema enforced during generation. At search time each sees the question, the object, the time, the rooms and their spots, how long since it last looked in each room, and what the rooms opened so far held. Each night it sees all of the day’s sightings and its current memory and writes the updated memory.

- *Last-seen* (no language model). Keeps, for each object, the rooms it has been seen in, newest first, and opens them in that order. An object never seen is searched where objects of the same kind were last seen.
- *Summary-only memories*. Three designs that keep only written text. The *claim store* is a list of claims about the household, each citing the sightings for and against it and carrying a condition under which it holds; the model edits individual claims. The *ACE-style play-book* adds, revises and merges claims through nightly reflection [3]. The *MemGPT-style working memory* is a fixed-size block of free text edited by naming a piece and its replacement, plus an archive the model must search [2]; we run a reduced 1,200-character variant. Implementation differences from the published systems are in Appendix D.
- *Log and notes*. The claim store, plus the raw log of the asked object at search time: its newest forty sightings, one summary line for older ones, and the rooms where searching did not find it.
- *Notes hidden at search time*. A control that writes the claim store but chooses rooms without reading it.

3 Results

3.1 A Simple Rule Outperforms the Summary-Only Memories

Table 1 gives accuracy over the whole 31-day run. Last-seen finds 95.0% of objects within three rooms. The summary-only memories find 82.1–86.9%. The last-seen lead ranges from 8 to 13 points on found within three rooms and holds in 10 of 10 households against the claim store. Hiding the notes at search time costs about 30 points, so the notes do carry information about where to look. The summary-only designs simply carry less than the raw record.

Log and notes scores highest. It shares the claim store’s nightly update, so the difference between them is only that log and notes can read the object’s sightings when choosing a room. That difference is worth 8.6 points on first room right and 9.7 on found within three rooms. The gain is not a better first guess: on questions where neither has yet seen the object in its new place, log and notes and

Method	First room right (%)	Found within 3 rooms (%)
Log and notes	85.6	96.6
Last-seen (no language model)	82.1	95.0
Claim store	77.0	86.9
ACE-style playbook	76.0	85.3
MemGPT-style working memory	71.0	82.1
Notes hidden at search time	53.5	65.8

Table 1: **Accuracy over all questions, ten households, 8 questions a day**, 2,480 questions per method. Last-seen leads every summary-only memory; each margin clears two standard errors across households and the noise floor on found within three rooms (Appendix A). Log and notes leads last-seen by +3.5 / +1.5 points, which does not clear the floor.

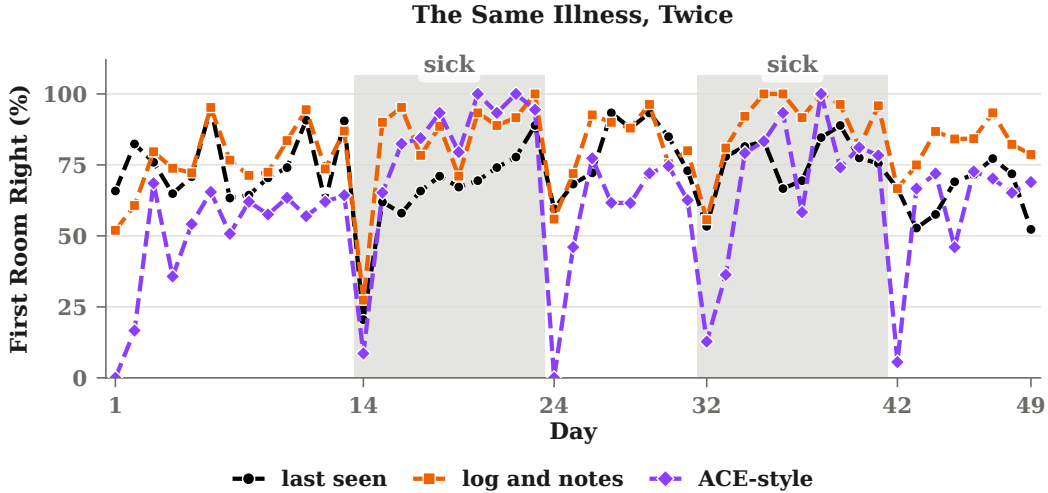


Figure 2: **The same illness, twice.** First room right by day on the 14 objects that change room in both illnesses, three households at 24 questions a day, illness days shaded. Every memory drops when a change begins. The two memories that keep raw sightings drop about half as far at the second illness as at the first. The claim store is omitted for clarity; it scored below the methods shown, and its values are in Table 2.

118 last-seen are level. The gain is what happens after a wrong first guess. On the first day of the illness
119 the two guess the first room about equally often on moved objects, but log and notes finds the object
120 within three rooms 57% of the time against last-seen’s 24%, because the model reasons about where
121 else the object could be while last-seen walks back through rooms the object used to occupy. One
122 day of this decides the week: by days 15–17 log and notes has seen the moved objects in their new
123 places on 95% of questions against 59% for last-seen. The advantage grows when the robot observes
124 less and shrinks when it observes more (Appendix C).

125 This matches the finding of Zhang et al. [5] on text-agent benchmarks that raw episodes are com-
126 petitive with consolidated memory. Here the record is not curated trajectories but the robot’s own
127 sparse observations, about ten room openings a day across a whole home.

128 3.2 When the Illness Returns, Only Memories With a Log Are Less Wrong

129 Figure 2 follows the three-household run through both illnesses. If experience from the first illness
130 helped, a memory should be less wrong on day 32 than it was on day 14. We compare the two days
131 on the 14 objects that change room in both illnesses.

132 Measured against each method’s own mean over the four preceding days, last-seen drops 59 points
133 in first room right on day 14 and 31 on day 32. Log and notes drops 57 and 29. The summary-only

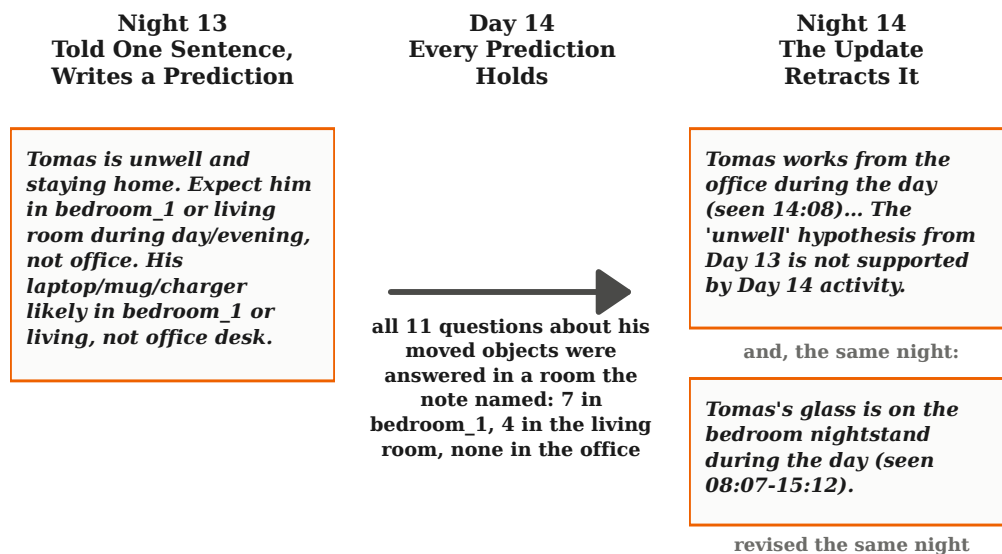


Figure 3: **Told the cause, the model predicts correctly and then retracts.** One household, log and notes, told one sentence on night 13 and asked what will change. All 11 questions about the resident’s moved objects the next day landed where the note said. The next update dropped the hypothesis on one office sighting while, in the same pass, recording the evidence for it. Quotes verbatim; the ellipsis marks one omitted sentence.

134 memories drop as far the second time as the first: the claim store 51 and 48, the ACE-style playbook
135 53 and 55. On found within three rooms, last-seen gains 57 points from day 14 to day 32 and log
136 and notes 29, while the claim store and playbook lose ground (Table 2).

137 The difference is what each memory still held from the first illness on night 31, a week after recovery.
138 Last-seen’s later guesses are its older rooms, and by day 32 those reach back to the first illness:
139 every one of its second- or third-room finds in a first-illness room came from a day-23 sighting. Log
140 and notes kept some first-illness placements as notes with a time-of-day condition, right room and
141 wrong reason, which was enough to make the room a fallback. The summary-only memories held
142 neither. On night 31 the claim store had no note placing any of the three ill residents’ glasses in
143 the bedroom and the playbook had one, and on day 32 neither opened a bedroom for those glasses.
144 Three households is a small sample; the two-standard-error bands in Table 2 are wide, and we read
145 this as the direction of the effect rather than its size.

146 3.3 The Memories Record Where Things Went, Never Why

147 We audited what the memories held on night 31 of the ten-household run. The claim store, log and
148 notes and the ACE-style playbook together hold 426 live notes that name an object with a room it
149 moved to during the illness. None mentions illness. Most attach a time of day; the rest say “always”
150 or name a person being home (Appendix B). The claim store has a field for the condition under
151 which a claim holds. Across six runs in which the model was told about the illness, 361 of 361
152 claims fill it and one names being unwell.

153 Figure 3 shows what the model can do when handed the cause. Told in one sentence on night 13
154 that a resident is unwell and asked what will change, it wrote a correct prediction before seeing a
155 single changed placement: his things would be in the bedroom or living room, not the office. Every
156 question about his moved objects the next day landed where the note said. This is what we hoped a

157 language model would bring: a rule cannot use a stated cause it has never observed, and the model
158 could.

159 The next night the model saw the resident in the office once and dropped the hypothesis. In the same
160 pass it revised another note to record his glass on the bedroom nightstand from 08:07 to 15:12. In
161 the normal routine that glass is never on the nightstand; through the illness it is there for 90% of the
162 working day. A model reasoning about the household would read that as the illness written down.
163 This one filed it as a standing fact about the glass, a few lines below the note where it deleted the
164 explanation.

165 The update, not the inference, is where this fails. Each night’s update reasons from that night’s
166 sightings alone, so a single contrary sighting retires a hypothesis while a confirming one is absorbed
167 as unrelated. A memory that works this way cannot accumulate evidence for an explanation across
168 days, which is why the summary-only memories are as wrong at the second illness as at the first:
169 one day never looks like a regime.

170 4 Discussion

171 Our results agree with Zhang et al. [5] on the central point: the raw record of experience is compet-
172 itive with, and here better than, the summaries a language model writes from it. They reach this in
173 static text-agent worlds where every episode is complete. We reach it in a simulated home where
174 observations are sparse, chosen by the memory itself, and never labeled with the cause of a change.
175 The failure survives the move from curated trajectories to a robot’s own partial searches.

176 Our setting also shows something a static world cannot. Zhang et al. attribute the degradation to
177 rewriting corrupting knowledge that should have been stable. In our households, some knowledge
178 is supposed to change. The memories did revise when the routine changed; every method adapted
179 within the first illness. What they lost was the conditional structure: that a set of placements held
180 under one circumstance and would hold again when it returned. The told-cause case locates this
181 loss in the update. The model held the right hypothesis and used it, then retired it on one contrary
182 sighting while filing the confirming sighting as an unrelated fact.

183 This changes what the raw record is for. In Zhang et al., keeping episodes protects against corruption.
184 Here, keeping sightings is what let two methods do better at the second illness, because the old
185 regime was still readable when it came back. It also changes what to evaluate. Robot memory
186 benchmarks measure recall within an episode [6, 7] or degradation as unrelated history grows [8].
187 Both are single-change measures, and a memory that only tracks recent observations passes them. A
188 change that returns is what separates a memory that learned from one that kept up, and we propose
189 it as a standard test: report each method’s drop at the second onset against its own drop at the first.

190 Three design directions follow. Keep the raw record alongside summaries. Represent the state of
191 the household, such as “a resident is unwell, since day 14,” separately from claims about objects, so
192 a claim tied to a state can be set aside and restored rather than rewritten. And gate the update so that
193 one contrary sighting cannot retire a hypothesis that a day of evidence supported.

194 **Limitations.** The homes are simulated and perception on opening a room is perfect, so this isolates
195 the memory problem from the perception problem. There is one kind of change and the second
196 illness affects the same resident, so we test reuse of a recurring distribution rather than transfer to
197 a new resident or a new kind of change. We use one language model with thinking disabled. The
198 recurrence result rests on three households and the told-cause result on one. Our ACE-style and
199 MemGPT-style memories are reduced variants of the published systems (Appendix D). An arm that
200 reads the log without notes would separate the value of the summaries from the model’s use of the
201 log; we did not run it.

5 Conclusion

A language model that writes its own memory was expected to out-search a simple rule when a household routine changed and returned. In this setting it did not. A rule that searches where each object was last seen outperformed the summary-only memories, and only memories that kept raw sightings were less wrong when the illness came back. The memories recorded where things went but never why, and a correct prediction from a stated cause was undone by the next update. Self-updating robot memories should be judged on what they preserve and reuse when a situation returns.

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A Setting Details and Statistics

Households. Day 0 is a night with no questions. Between 10 and 20 of the 80 to 100 objects in each household are asked about, to focus on the effects of the change in routine. Questions cover eleven kinds of object, including books, mugs, glasses, water bottles, chargers and tablets. The question schedule depends only on the question bank and is identical for every method. The set of moved objects is the same for every method.

244 **Observation budget.** The number of questions per day sets how many rooms the robot observes:
 245 at 8 a day it opens 9.0–10.7 rooms, at 24 a day 25.6–30.5.

246 **Statistics.** We compare methods by paired differences within each household, averaged across
 247 households, and report two standard errors of that average. Generation at temperature zero is not
 248 exactly repeatable on our server, so we measured a noise floor from two arms whose prompts are
 249 byte-identical over the settled days of a separate run: 1.9 points on first room right and 2.2 on
 250 found within three rooms. We call a difference clear when it exceeds both and has the same sign in
 251 every household. Questions within a household are not independent, so the household is the unit of
 252 comparison.

Method	Home A	Home B	Home C	Mean change	2 SE
<i>Found within three rooms</i>					
Last-seen (no language model)	10 → 78	33 → 100	27 → 64	+56.9	20.6
Log and notes	40 → 78	50 → 100	82 → 82	+29.3	30.1
Claim store	20 → 33	50 → 25	73 → 36	−16.0	30.1
ACE-style playbook	10 → 44	83 → 25	91 → 45	−23.1	58.0
<i>First room right</i>					
Last-seen (no language model)	10 → 56	33 → 50	18 → 55	+32.9	17.0
Log and notes	20 → 44	17 → 50	45 → 73	+28.4	5.2
Claim store	10 → 11	0 → 0	9 → 27	+6.4	11.8
ACE-style playbook	0 → 11	17 → 0	9 → 27	+4.2	21.3
Questions, day 14 → 32	10 → 9	6 → 4	11 → 11		

Table 2: **First day of each illness, day 14 → day 32**, on the 14 objects that change room in both illnesses. Three households, 24 questions a day. Percentages; mean change and two standard errors from paired household differences. Objects never found: last-seen 21 → 6, log and notes 11 → 4, claim store 14 → 16, ACE-style 11 → 14.

Method	Home A	Home B	Home C	Mean change	2 SE
Last-seen (no language model)	25 → 85	64 → 100	27 → 67	+45.1	14.6
Log and notes	50 → 85	73 → 100	82 → 83	+21.1	20.1
Claim store	33 → 54	64 → 33	73 → 42	−13.6	34.1
ACE-style playbook	25 → 62	91 → 17	91 → 50	−26.2	65.6
Questions, day 14 → 32	12 → 13	11 → 6	11 → 12		

Table 3: **Found within three rooms, day 14 → day 32, on all 22 objects that moved in the first illness.** Eight of these do not change room in the second illness, which makes day 32 easier for every method. The pattern matches Table 2.

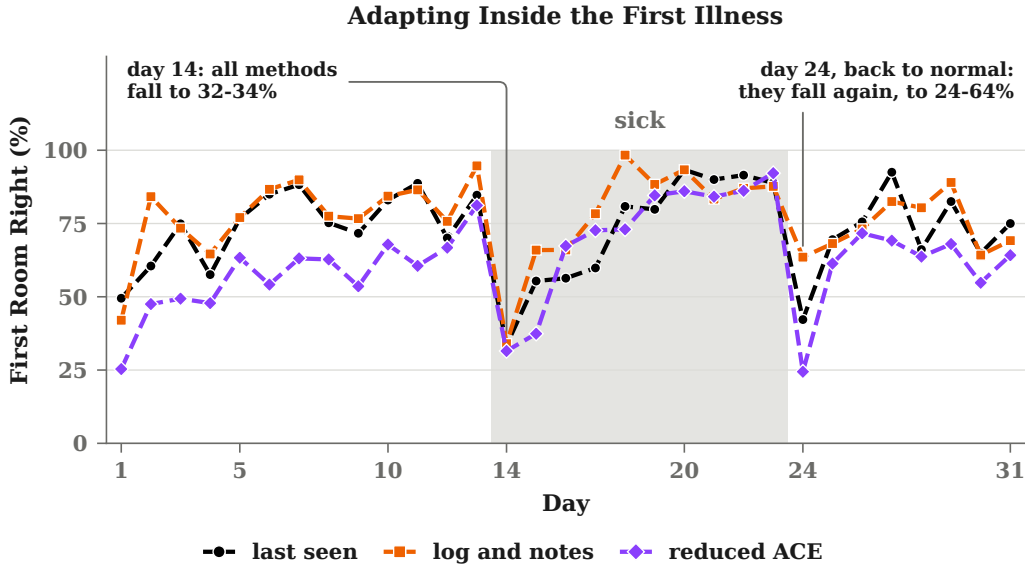


Figure 4: **The first illness, day by day.** First room right on the objects the illness moves, ten households at 8 questions a day; illness days shaded. Every memory drops to about a third on day 14 and climbs back within the illness. The drop on day 24, when normal life returns, is similarly deep.

253 B What the Notes Held

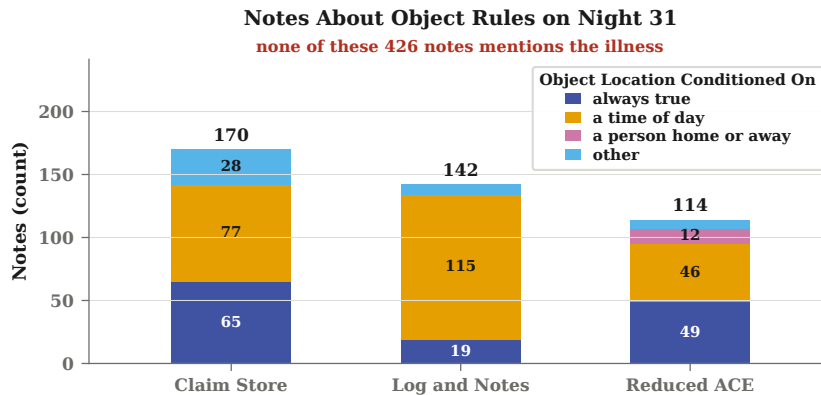


Figure 5: **Where things went, never why.** Notes held on night 31, a week after recovery, that place an object where it went during the illness, by the condition the note attaches; claim store, log and notes and ACE-style playbook, ten households. Most attach a time of day, which is true again the next day, ill or well. MemGPT-style is left out: 101 notes against the claim store's 595, too few for its bar to say anything.

254 C Showing the Sightings at Search Time Improves the Claim Store

255 The claim store and log and notes share the same nightly update; the one change is that log and
 256 notes also sees the object's raw sightings when choosing where to look. Because the two arms open
 257 different rooms, they also go on to see different things, so the +8.6 / +9.7 figure in Section 3.1
 258 includes what the different searches then collected.

259 Seeing is not enough on its own. By days 15–17 the claim store has seen the moved objects in their
260 new places on 96% of questions and still finds them only 62% of the time, because it rewrote those
261 sightings into summaries it cannot read back.

262 How much the summaries add depends on how much the robot observes. On the five other house-
263 holds, log and notes leads last-seen by +4.4 and +5.0 points at 4 questions a day, in all five house-
264 holds, by +3.5 and +2.6 at 8, and by +1.2 and +0.1 at 24. The advantage is recovery from a wrong
265 first guess, so it grows when the robot gets fewer looks a day and each one has to count; with plenty
266 of looks the record alone is enough.

267 **D Implementation Details**

268 **Differences from ACE.** Our implementation keeps ACE’s three-role division (a generator, a re-
269 flector and a curator) but differs from the released code in several ways. The released reflector loops
270 up to three rounds and only when the answer was wrong; ours runs unconditionally each night, once
271 in the ten-household run and up to three times in the 50-day run. The released grow-and-refine step
272 groups bullets by sentence embedding at cosine 0.90 and has the model author the merged text; ours
273 ranks pairs by embedding, hands at most four pairs a night to the model, which decides whether they
274 say one thing, and merges by deterministic concatenation. Pairs are restricted to notes holding under
275 the same condition. The released code defines add, update, merge and delete but executes only add;
276 ours runs revise, attach-evidence and join, and never deletes. The condition field on each claim is
277 our addition and is not part of ACE. The generator here is the robot’s own room choices, and credit
278 is whether the first room opened held the object.

279 **Differences from MemGPT.** The working memory is one block of free text, edited by naming an
280 exact piece of it and its replacement; a piece that is not present fails, as in MemGPT. Passages can
281 be inserted into an archive and the archive searched a page at a time; a write that would overflow
282 the block is refused. The ten-household run used a 1,200-character block, a sixteenth of MemGPT’s
283 20,000, and is reported as a reduced variant. A 20,000-character version ran only on three house-
284 holds at 24 questions a day for 31 days and is not reported here. The 50-day run has no MemGPT-
285 style arm.

286 **The log window.** The log shows the newest forty sightings of the asked object; at 24 questions a
287 day the cap binds on most questions. It does not cost accuracy: log and notes’ lead over last-seen is
288 no smaller on questions where the cap bit, accuracy does not step down where the dated lines stop,
289 and on questions the cap never touched the lead still falls from +4.4 at 4 questions a day to +0.0 at
290 24. Whether a larger window would help is untested.

291 **Room-choice prompt.** In the ten-household run the room-choice prompt named the room before
292 its reason; a paired check found this does not move the day curves.